

Study Materials Order for DS/ML Placement

Follow this order exactly. The goal is placement readiness, not collecting unlimited courses.

Main target: Data Scientist Intern, ML Engineer Intern, Applied AI Intern, or Data Analyst with ML.

Rule: Study one phase, build the required output, then move to the next phase. Do not jump to RAG before revising ML and SQL.

Complete Order

Order	Phase	Time	Main Output
1	ML Revision and Interview Depth	10 days	ML interview notes + improved Bank Churn explanation.
2	SQL for Placement	15 days	SQL practice file/repo with 40 to 60 solved queries.
3	NLP Fundamentals	15 days	Text classification project using TF-IDF and ML.
4	LLM, Embeddings, and RAG	20 days	RAG-based Resume and JD Matcher or Chat with PDF app.
5	Deployment	10 days	Streamlit demo + FastAPI prediction endpoint.
6	Resume, GitHub, and Interview Practice	Ongoing	Placement-ready resume, pinned repos, mock answers.

Phase 1: ML Revision and Interview Depth

Goal: Revise classical ML from interview perspective. You already know ML, so do not restart a full beginner course.

Use These Materials In Order

1. Google Machine Learning Crash Course

<https://developers.google.com/machine-learning/crash-course>

Study only the important ML concepts: supervised learning, generalization, overfitting, classification, regularization, and evaluation.

2. Scikit-learn Supervised Learning Guide

https://scikit-learn.org/stable/supervised_learning.html

Use this as a reference for Logistic Regression, SVM, Naive Bayes, Decision Trees, Random Forest, and Gradient Boosting.

3. Scikit-learn Model Selection and Evaluation

https://scikit-learn.org/stable/model_selection.html

Study train-test split, cross-validation, grid search, and metrics.

Topics To Finish

- Regression vs classification.
- Linear Regression, Logistic Regression, KNN, SVM, Naive Bayes.

- Decision Tree, Random Forest, Gradient Boosting, XGBoost concept.
- KMeans, DBSCAN, hierarchical clustering basics.
- Train-test split, cross-validation, hyperparameter tuning.
- Accuracy, precision, recall, F1, ROC-AUC, PR-AUC, confusion matrix.
- Overfitting, underfitting, bias-variance, regularization.
- Data leakage, imbalanced data, pipelines.

Output Before Moving On

- Write 20 ML interview Q&A from your Bank Churn project.
- Prepare a 2-minute explanation of your Bank Churn project.
- Update README with model metrics and why XGBoost was selected.

Phase 2: SQL for Placement

Goal: Become comfortable with SQL questions asked in data roles and campus placement rounds.

Use These Materials In Order

1. SQLBolt

<https://sqlbolt.com/>

Complete every lesson. This is interactive and very good for fundamentals.

2. Kaggle Learn

<https://www.kaggle.com/learn>

Do Intro to SQL and Advanced SQL. Also do Pandas and Data Cleaning if you need revision.

Topics To Finish

- `SELECT` , `WHERE` , `ORDER BY` , `LIMIT` .
- `GROUP BY` , `HAVING` , aggregations.
- Inner join, left join, right join, full outer join concept.
- Subqueries and CTEs.
- Window functions: `ROW_NUMBER` , `RANK` , `DENSE_RANK` , `LAG` , `LEAD` .
- Case statements and conditional aggregation.
- Primary key, foreign key, normalization, star schema basics.

Output Before Moving On

- Create a file or repo named `SQL-Interview-Practice` .
- Solve 40 to 60 queries.
- Add at least 10 business queries from your own dashboard projects.

Phase 3: NLP Fundamentals

Goal: Cover the NLP gap visible in many ML/AI job descriptions.

Use These Materials In Order

1. **Scikit-learn Text Feature Extraction**

https://scikit-learn.org/stable/modules/feature_extraction.html#text-feature-extraction

Learn Bag of Words, CountVectorizer, TF-IDF, and text classification workflow.

2. **Hugging Face NLP/LLM Course**

<https://huggingface.co/learn/nlp-course>

Start after TF-IDF basics. Focus on transformer tasks, tokenizers, datasets, and pipelines.

Topics To Finish

- Text cleaning: lowercase, punctuation, stopwords.
- Tokenization.
- Stemming and lemmatization concept.
- Bag of Words and n-grams.
- TF-IDF.
- Sentiment analysis and text classification.
- Named Entity Recognition basics.
- Embeddings concept.

Output Before Moving On

- Build a Customer Review Sentiment Analyzer.
- Use TF-IDF plus Logistic Regression or Naive Bayes first.
- Compare at least 2 models.
- Evaluate using accuracy, precision, recall, F1, and confusion matrix.
- Create a Streamlit demo later in Phase 5.

Phase 4: LLM, Embeddings, and RAG

Goal: Build practical RAG knowledge for modern Applied AI roles. Learn enough to build and explain one project.

Use These Materials In Order

1. **LangChain RAG Tutorial**

<https://docs.langchain.com/oss/python/langchain/rag>

Follow this to understand loading, splitting, storing, retrieving, and generating.

2. **LangChain Retrieval Docs**

<https://docs.langchain.com/oss/python/langchain/retrieval>

Study retrieval concepts after the basic RAG tutorial.

3. **FAISS Documentation**

<https://faiss.ai/>

Use this to understand vector similarity search basics.

Topics To Finish

- LLM basics: tokens, context window, temperature.
- Prompt engineering basics.
- Embeddings and semantic similarity.
- Vector search and vector databases.
- Chunking documents.
- RAG pipeline: load, split, embed, store, retrieve, generate.
- Hallucination and source grounding.
- Fine-tuning vs RAG.
- Basic agent/tool-calling concept, optional.

Output Before Moving On

- Build a Resume and JD Matcher or Chat with PDF app.
- Use embeddings and vector search.
- Return a structured answer: match score, matching skills, missing skills, project suggestions.
- Add a README explaining what RAG is and why your app uses it.

Phase 5: Deployment

Goal: Convert notebook projects into usable apps and APIs.

Use These Materials In Order

1. **Streamlit Documentation**
<https://docs.streamlit.io/>
Use Streamlit for demos. It is the fastest way to show ML/NLP/RAG apps.
2. **Streamlit Installation Guide**
<https://docs.streamlit.io/get-started/installation>
3. **FastAPI Official Tutorial**
<https://fastapi.tiangolo.com/tutorial/>
Use FastAPI for model prediction endpoints.

Topics To Finish

- Streamlit inputs, buttons, forms, charts, file upload.
- Saving ML models using `joblib` or `pickle`.
- FastAPI `GET` and `POST` endpoints.
- Create a `/predict` endpoint for an ML model.
- `requirements.txt` and project structure.
- README with setup instructions and screenshots.

Output Before Moving On

- Deploy or locally demo Bank Churn with Streamlit.
- Create a FastAPI endpoint for one model.
- Deploy NLP or RAG project with Streamlit if possible.

Phase 6: Resume, GitHub, and Interview Practice

Goal: Make your work visible and explainable to recruiters and interviewers.

What To Do

- Pin Bank Churn, NLP project, RAG project, and one dashboard project.
- Add repo descriptions for all important projects.
- Use README sections: problem, tech stack, workflow, result, screenshots, how to run.
- Prepare 2-minute explanation for each project.
- Practice SQL and DSA daily.

Interview Practice Topics

- Why recall matters in churn prediction.
- ROC-AUC vs PR-AUC.
- How pipelines prevent leakage.
- How TF-IDF converts text to numbers.
- What embeddings are.
- What RAG is and why it reduces hallucination.
- How your app is deployed.
- SQL joins, CTEs, and window functions.

Weekly Study Routine

Task	Daily Time	Why
SQL	45 minutes	High chance of appearing in data roles.
DSA / Python Coding	45 minutes	Useful for campus placement filters.
Current Phase Study	60 minutes	Build theory in the right order.
Project Work	90 minutes	Turns learning into portfolio proof.
Notes / README	15 minutes	Makes interview revision easier.

Things To Avoid For Now

These are useful later, but not your current priority for placement.

- Another complete beginner ML course.
- Deep learning from scratch in heavy detail.
- Advanced robotics, CUDA, Rust, C++ optimization.
- Advanced cloud architecture.
- Deep fine-tuning of Llama/Mistral.

- Too many certificates without projects.

Final Checklist

- ML revision completed.
- 40 to 60 SQL questions solved.
- NLP text classification project completed.
- RAG project completed.
- At least one Streamlit app completed.
- At least one FastAPI prediction endpoint completed.
- GitHub pinned projects cleaned.
- Resume updated with metrics and project outcomes.